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Attitude control for hovering rocket decoy based on PSO-optimized sliding mode controller

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Abstract: Addressing the challenge of rapid attitude stabilization for hovering rockets subject to compound disturbances (including aerodynamic interference and thrust misalignment), this paper proposes a Particle Swarm Optimization (PSO)-enhanced Non-singular Fast Terminal Sliding Mode Control (NFTSMC) strategy. The rocket attitude dynamics model is first established. Subsequently, a non-singular fast terminal sliding surface is formulated, coupled with a switching control law utilizing the Adaptive Super-Twisting Algorithm (ASTA) to mitigate chattering. An Extended State Observer (ESO) is constructed for the real-time estimation of disturbances. Global stability of the closed-loop system is rigorously proven via Lyapunov analysis. Furthermore, PSO is leveraged for the global parametric optimization of the sliding mode controller, thereby improving control precision. Comprehensive simulation experiments validate the controller design and demonstrate its performance advantages.

1. Introduction

Rocket attitude control plays a critical role in aerospace engineering, particularly for specialized vehicles like hovering rocket decoys that demand rapid response, precise command-tracking capabilities, and robustness against strong disturbances. The hovering rocket decoys need to maintain a precise attitude during the operation to achieve electromagnetic suppression or thermal decoy function. Due to the absence of aerodynamic surfaces, low flight speed, and negligible aerodynamic forces, the hovering rocket body exhibits significant static instability. Thrust vectoring is adopted for attitude control of this type of hovering rocket. However, its flight environment is fraught with uncertainties: multi-source disturbances, including pitch-yaw channel control coupling and aerodynamic interference, make its attitude control a typical nonlinear time-varying problem. Traditional linear control methods (e.g., PID [1]) perform well under the assumption of fixed model parameters, but their control accuracy and robustness significantly degrade when confronted with the aforementioned strong disturbances and parameter perturbations. Therefore, designing a robust control architecture capable of real-time disturbance estimation and compensation is crucial for improving the mission reliability of the hovering rocket decoy.

This paper designs a Non-singular Fast Terminal Sliding Surface and a switching control law based on ASTA, develops a sliding mode controller combined with ESO, proposes a sliding mode parameter optimization strategy driven by the PSO algorithm, and verifies the control effectiveness through attitude tracking simulations.

2. System model description

The hovering rocket decoy generates lateral moments through the deflection of the gas rudder to alter the flight attitude of the body, thereby achieving attitude tracking. The body involves three channels during flight: pitch, yaw, and roll. Due to its low flight speed and small aerodynamic forces, the influence of body roll on motion is neglected in this paper to simplify the process.

Considering the symmetry of the body, the pitch-yaw channel dynamics are given by:

$$\mathbf{J}\ddot{\boldsymbol{\theta}} + \mathbf{D}\dot{\boldsymbol{\theta}} = \mathbf{M}_{\mathbf{gas}} + \mathbf{d} \tag{1}$$

where $\boldsymbol{\theta} = [\theta_{\mathrm{y}}, \theta_{\mathrm{p}}]$ is the attitude angle, $\mathbf{J} = [J_{\mathrm{y}}, J_{\mathrm{p}}]$ is the moment of inertia, $\mathbf{D} = [D_{\mathrm{y}}, D_{\mathrm{p}}]$ is the damping coefficient, $\mathbf{M}_{\mathbf{gas}} = [M_{\mathrm{y}}, M_{\mathrm{p}}]$ is the gas rudder control moment, and $\mathbf{d} = [d_{\mathrm{y}}, d_{\mathrm{p}}]$ is the compound disturbance (including wind disturbance, aerodynamic moments, etc.).

3. Controller design

3.1. Controller objective

To intuitively reflect the motion state of the body, the state where the body is hovering vertically upward is taken as the reference, i.e. $\boldsymbol{\theta} = [0,0]$ designs M_{gas} such that the actual attitude angle $\boldsymbol{\theta}$ tracks the command $\boldsymbol{\theta}_{\mathbf{d}} = [\sin t, \cos t]$ from different initial angles, achieving rapid convergence tracking, reducing system chattering, and satisfying robustness: suppressing compound disturbances $\mathbf{d}$ and parameter uncertainties $\boldsymbol{\Delta}\mathbf{J}$.

Due to the symmetry and consistency between the pitch and yaw channels, the design section uses the yaw controller design as a reference, meaning design parameters are presented in scalar form, while the actual pitch and yaw channels are controlled independently.

3.2. Non-singular fast terminal sliding surface

For the second-order system attitude control model of the hovering rocket:

$$\begin{cases} \dot{e}_1 = e_2 \\ \dot{e}_2 = \dfrac{1}{J}\left(M_{\mathrm{y}} - D\dot{\theta} + d\right) \end{cases} \tag{2}$$

where $e_1 = \theta_{\mathrm{d}} - \theta$ and $e_2 = \dot{\theta}_{\mathrm{d}} - \dot{\theta}$ are the system state variables.

The sliding surface is designed [2-3]:

$$s = e_2 + \alpha e_1 + \beta |e_1|^{\gamma} \operatorname{sgn}(e_1) \tag{3}$$

where $\alpha > 0$, $\beta > 0$, and $\gamma = p/q$ are sliding surface coefficients; p and q are positive odd integers satisfying $1 \le p/q \le 2$.

Differentiating s and substituting $\dot{e}_2 = \ddot{\theta}_{\mathrm{d}} - \ddot{\theta}$ yields:

$$\dot{s} = \ddot{\theta}_{\mathrm{d}} - \ddot{\theta} + \alpha e_2 + \beta\gamma e_2 |e_1|^{\gamma-1} \operatorname{sgn}(e_1) \tag{4}$$

Given $\dot{s} = 0$, the equivalent control input is:

$$u_{\mathrm{eq}} = \ddot{\theta}_{\mathrm{d}} + \alpha e_2 + \beta\gamma e_2 |e_1|^{\gamma-1} \operatorname{sgn}(e_1) \tag{5}$$

Analysis of $\dot{s}$ shows that when $\gamma > 0$, $|e_1|^{\gamma-1}$ is bounded at $e_1 = 0$, avoiding singularity. When the control system state is relatively close to the equilibrium point $e_1 \to 0$, the $\beta\gamma e_2 |e_1|^{\gamma-1} \operatorname{sgn}(e_1)$ term is larger; when the control system state is far from the equilibrium point, the αe_2 term is larger. In summary, this sliding surface enables global convergence.

3.3. Adaptive switching term

Traditional sliding mode control switching terms often use the sign function, leading to chattering in the control input during high-frequency switching. This chattering can damage actuators, increase energy consumption, and reduce control accuracy. The Super-Twisting Algorithm (STA) [3] replaces the sign function with a nonlinear integral term, generating a continuous and smooth control signal to handle high-frequency unmodeled dynamics. Its switching control law form is:

$$\begin{cases} u = -k_1 \, |s|^{1/2} \, \mathrm{sign}(s) + v \\ \dot{v} = -k_2 \mathrm{sign}(s) \end{cases} \tag{6}$$

where s is the sliding surface; k_1, k_2 is the switching term coefficients. The differential term $\dot{v}$ can estimate and compensate for the system's lumped disturbances in real time, enabling precise tracking of the sliding surface under disturbances and enhancing robustness.

This paper designs a switching control law based on the ASTA [4], with its parameter structure given by:

$$\begin{cases} k_1 = k_{1,\mathrm{base}} \left(1 + p \tanh(q\,|s|)\right) \\ k_2 = k_{2,\mathrm{base}} \left(1 + r \left|\tanh(I_s)\right|\right) \end{cases} \tag{7}$$

where $k_{1,\mathrm{base}}, k_{2,\mathrm{base}}$ is the base adjustment parameters; p, q, r is the adaptive gain; p is the modulus adjustment factor, achieving smooth switching between high gain in large-error regions and low gain in small-error regions via the hyperbolic tangent function; q is the boundary layer sensitivity regulator, controlling the gain change rate through the attenuation characteristics of $sech^2(q\,|s|)$ (when q increases, $sech^2(q\,|s|)$ decays rapidly in the $|s| > 0$ region, making gain adjustment more sensitive near the sliding surface); r is the integral term disturbance rejection enhancer, where $I_s = \int \tanh(s/\delta) dt$, $\delta = 0.1(1 - e^{-0.1|s|})$ .When persistent disturbances cause $\left|I_s\right|$ to increase, r enhances the k_2 gain to strengthen the disturbance rejection capability.

The adaptive gain dynamic equation consists of triply coupled parameters:

$$\begin{cases} \dot{p} = -\chi_p \, |s| \tanh(q\,|s|) \\ \dot{q} = -\chi_q p \, |s| \, \mathrm{sech}^2(q\,|s|) \\ \dot{r} = \chi_r \, |I_s| \, \mathrm{sech}^2(q\,|s|) \end{cases} \tag{8}$$

where χ_p, χ_q, χ_r is the gain learning rate; q regulates the response speed of p ; p regulates the decay rate of q . By linking the integral term amplitude to the gain via $\dot{r} \propto \left|I_s\right|$, dynamic compensation for step/ramp disturbances is achieved.

3.4. Disturbance observer

A linear disturbance observer is designed to achieve unified estimation of matched/unmatched low-frequency disturbances by extending the disturbance as a system state [5]. The objective is to estimate the full state vector through the output θ.The ESO structure dynamic equation is designed as:

$$\begin{cases} \dot{z}_1 = z_2 + w_1(\theta - z_1) \\ \dot{z}_2 = z_3 + w_2(\theta - z_1) + J^{-1} M_{\mathrm{y}} \\ \dot{z}_3 = w_3(\theta - z_1) \end{cases} \tag{9}$$

where z_1 is the angle estimate, z_2 is the angular rate estimate, and z_3 is the disturbance estimate. According to the empirical formula, the gain meets $w_1 = 3\boldsymbol{\omega}_0$, $w_2 = 3\boldsymbol{\omega}_0^2$, and $w_3 = \boldsymbol{\omega}_0^3$ (e.g., $\boldsymbol{\omega}_0 = (3 \sim 5)\boldsymbol{\omega}_{\mathrm{c}}$, where $\boldsymbol{\omega}_{\mathrm{c}} \approx \alpha$ = controller bandwidth). Setting $\boldsymbol{\omega}_0 = 3\boldsymbol{\omega}_{\mathrm{c}}$, dynamic gains $w_1 = \min(3\boldsymbol{\omega}_0 + 5\,|e_1|, 3\boldsymbol{\omega}_0 + 20)$ and $w_2 = \min(3\boldsymbol{\omega}_0^2 + 50\,|e_1|, 3\boldsymbol{\omega}_0^2 + 300)$. ESO responds rapidly to large errors and remains stable for small errors. The observed disturbance $\hat{d} = z_3$ enables dynamic cancellation.

3.5. Particle swarm parameter optimization (PSO)

The PSO algorithm iteratively updates the position and velocity of the particle swarm to gradually approach the optimal solution [6-7]. Its core formulas are as follows:

$$\begin{gathered} v_i^{t+1} = \omega v_i^t + c_1 r_1^t \left[p_i^t - x_i^t \right] + c_2 r_2^t \left[g_i^t - x_i^t \right] \\ x_i^{t+1} = x_i^t + v_i^{t+1} \end{gathered} \tag{10}$$

where ω is the inertia weight, balancing global exploration and local exploitation capabilities. During the initial stages of the algorithm execution, a larger inertia weight ($\omega_{\max}$) is employed to enhance global exploration for the optimal solution. As the iteration process proceeds, inertia weight linearly decreased to a smaller value ($\omega_{\min}$) to emphasize local exploitation; v_i is the particle velocity; x_i is the particle position, continuously updated with the number of iterations; c_1, c_2 are the cognitive and social learning factors, respectively adjusting the influence of individual experience and group collaboration; r_1, r_2 are random numbers between $[0, 1]$, enhancing search randomness; t is the iteration number; p_i is the individual best position; g_i is the global best position. The particle position must be constrained within the feasible domain, i.e., $x_i \in [\mathrm{lb}, \mathrm{ub}]$, to avoid system instability caused by parameter boundary violations.

When c_1 is set to a relatively large value, particles are strongly attracted back to their individual historical best positions. This facilitates local exploitation and helps maintain individual diversity within the swarm. However, it consequently slows down the convergence speed. Conversely, a smaller c_1 allows particles to respond more rapidly to the swarm's global best solution, but diminishes local exploitation capabilities. A relatively large c_2 strongly attracts particles towards the swarm's best solution, thereby increasing susceptibility to premature convergence into local optima. In contrast, a smaller c_2 reduces the influence of the swarm's best position on particles. This enhances global exploration capabilities, but the reduced inter-particle communication leads to a significant decline in convergence speed. In standard PSO implementations, Parameters c_1 and c_2 are typically set to 2.0 or undergo minor tuning within the range 1.8 to 2.0. This standard configuration generally exerts minimal adverse impact on the search for the optimal solution.

The design optimization goal is to minimize the tracking error, and the fitness function design takes the attitude tracking MSE as the optimization goal:

$$H(x) = \frac{1}{N} \sum_{k=1}^{N} (\theta_{\mathrm{d}}(k) - \theta(k))^2 \tag{11}$$

$$\lim_{t \to \infty} g_i(t) = x^*, \quad H(x^*) \le H(x_i) \quad \forall_i \tag{12}$$

where x^* is the global optimal solution. Considering the characteristics of the sliding surface, this paper optimizes the sliding surface coefficients α, β using the PSO algorithm, mapping the particle position x^* to the coefficients.

4. Stability proof

The control law designed based on the by system model is:

$$M_{gas} = J(u_{\mathrm{eq}} - u_{\mathrm{sw}}) + D\dot{\theta} - \hat{d} \tag{13}$$

where u_{eq} is the equivalent control term, u_{sw} is the switching control term, and $\hat{d} = z_3$ is the disturbance compensation term.

Differentiating the sliding surface s and substituting the control law yields:

$$\begin{cases} \dot{s} = -k_1 |s|^{1/2} \operatorname{sgn}(s) + k_2 z + \eta \\ \dot{z} = -\operatorname{sgn}(s) \end{cases} \tag{14}$$

where $k_1 > 0$, $k_2 > 0$, and $\eta = J^{-1}(d - \hat{d})$. It is assumed that bounded disturbances $|\eta| \le \overline{\eta}$.

We select a positive definite function to construct the Lyapunov function:

$$V(s,z) = |s| + \frac{1}{2}k_2 z^2 \tag{15}$$

This function is positive definite with respect to both s and z, and $s \to \infty$, $z \to \infty$ when $V(s,z) \to \infty$; $s = 0, z = 0$ if and only if $V(s,z) = 0$.

Differentiating the constructed Lyapunov function and substituting the control law yields:

$$\dot{V} = -k_1 |s|^{1/2} + \eta \operatorname{sgn}(s) \tag{16}$$

When $|s| > (\overline{\eta}/k_1)^2$, we have $-k_1 |s|^{1/2} + \overline{\eta} < 0 \Rightarrow \dot{V} < 0$, meaning $\dot{V}$ is negative definite. It can thus be concluded that the system state tends towards the origin, and the system will converge in finite time to a bounded region (sliding mode band) containing the equilibrium point: $|s| \le (\overline{\eta}/k_1)^2$, and remain bounded within it.

5. Simulation and analysis

During combat operations, the hovering rocket decoy is launched from the ground at various attitudes and follows command signals to reach designated airspace, or executes rapid maneuvers in flight by tracking commands. This demands both fast convergence speed for initial angular errors and precise tracking capability for diverse command profiles.

In practice, there exists coupling between pitch and yaw control for the hovering rocket, with a coupling coefficient magnitude of 10%-20% [1], along with random disturbances. A joint pitch-yaw control loop model is constructed, as shown in Figure 1.

The MATLAB Simulink toolbox is used to build a joint pitch-yaw control loop simulation system to verify the controller performance within the simulation environment.

Compound disturbances as shown in Figure 2 are added to the simulation system to simulate realistic wind and noise interference.

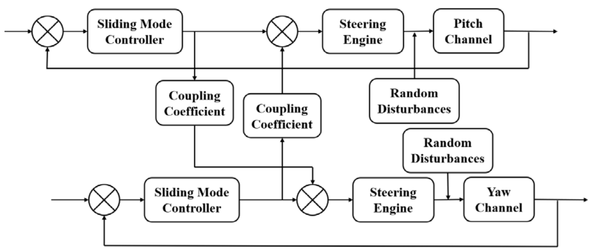

Figure 1. Joint pitch-yaw control loop.

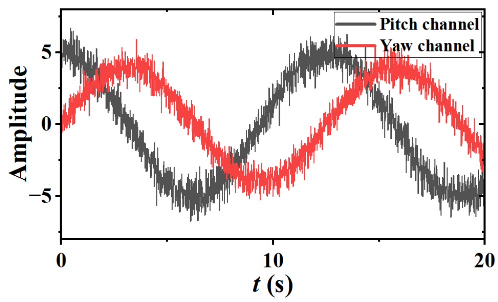

Figure 2. Composite interference.

The fundamental control parameters for pitch and yaw control are identical. Physical parameters: $J = 12\ \mathrm{kg}\cdot\mathrm{m}^2$, $D = 0.5$; Sliding mode controller parameters: α, β are parameters to be optimized, $\gamma = 5/3$; PSO parameters: $\omega_{\max} = 0.9$, $\omega_{\min} = 0.4$, $c_1 = c_2 = 2$, $\mathrm{lb} = 0.1$, $\mathrm{ub} = 10$; Number of particles $N = 100$; Maximum iterations 500; Switching term parameters: $k_{1,\mathrm{base}} = 1$, $k_{2,\mathrm{base}} = 0.1$, $\chi_p = 0.2$, $\chi_q = 0.3$, $\chi_r = 0.1$, $p \in [0.1, 2]$, $q \in [0.5, 3]$, $r \in [0.01, 0.5]$. The initial value of p, q is the midpoint of the interval. Since the initial integral term cannot be too large, $r = 0.1$ is taken. During actual control, due to physical parameter limitations of the gas rudder system, the control magnitude and its rate of change are constrained in the simulation: $M_\mathrm{y} \in [-60, 60]\ \mathrm{N}\cdot\mathrm{m}$, $\dot{M}_\mathrm{y} \in [-500, 500]\ \mathrm{N}\cdot\mathrm{m/s}$.

For the yaw angle, initial angles are set to $-60°$, $-30°$, $0°$, $30°$, and $60°$, and the pitch angle is set to $0°$. The coupling coefficient is $C = 0.15$, i.e., 15% of the moment in the pitch channel is injected into the yaw channel as cross-interference. We track the command $\boldsymbol{\theta}_\mathrm{d} = [\sin t, \cos t]$, with a simulation duration of 10 s. The sliding surface parameters of the yaw channel after PSO optimization and the convergence characteristics of the initial angle of 60° under different parameters are shown in Table 1.

Table 1. β and $H(x^*)$ after PSO Optimization.

θ_0 (°)	α	β	$H(x^*)$
-60	3.69	1.21	0.1571
-30	5.44	1.13	0.0375
0	2.15	3.63	0.0004
30	8.96	4.29	0.0095
60	8.11	3.34	0.0556

Table 2. The convergence characteristics of $60°$.

α	β	t (s)	$H(x^*)$
10.00	3.00	0.99	0.0626
8.00	5.00	1.88	0.0707
8.00	3.00	1.18	0.0565
8.00	1.00	1.05	0.0589
5.00	3.00	1.96	0.0684

For the initial angle $60°$, the PSO fitness optimized surface is shown in Figure 3, and the adaptive parameter change curves are shown in Figure 4.

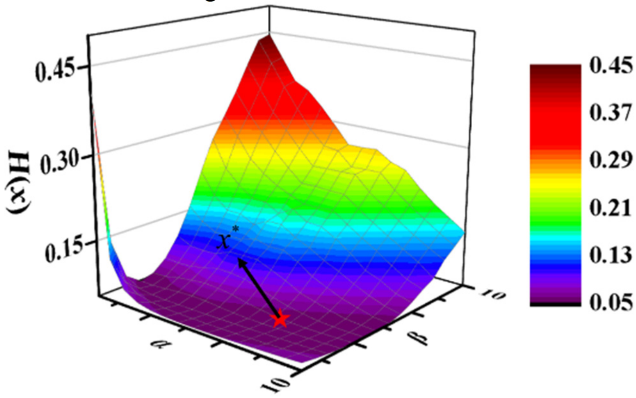

Figure 3. PSO fitness optimized surface.

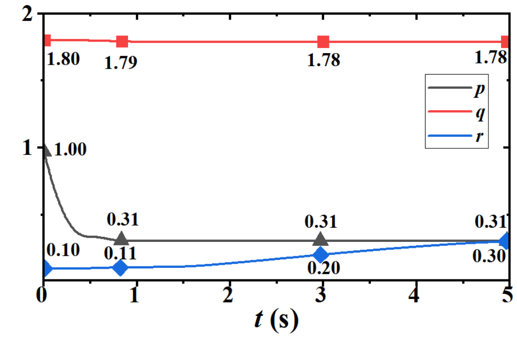

Figure 4. Adaptive parameter changing curves.

For the initial angle $60°$, analysis of Figure 3 shows that after PSO optimization, the sliding surface coefficients α,β are selected based on the coordinates of the point x^* with the best fitness. Combined with the analysis in Table 2, it can be seen that increasing α or decreasing β can improve the convergence speed of the system, but will lead to the decrease of the steady-state accuracy of the system. Analysis of Figure 4 shows that p,q gradually decrease during the convergence process. When the error approaches zero, p,q approximate a constant value. Due to the presence of persistent disturbances, $|I_s|$ increases, and r gradually increases accordingly, enhancing the compensation effect of the integral term.

The effect of different interference and coupling factor amplitudes on the system response at an initial angle of 60° is shown in Figure 5. Low-frequency disturbances are added to the system, and the compensation effect of the ESO is shown in Figure 6.

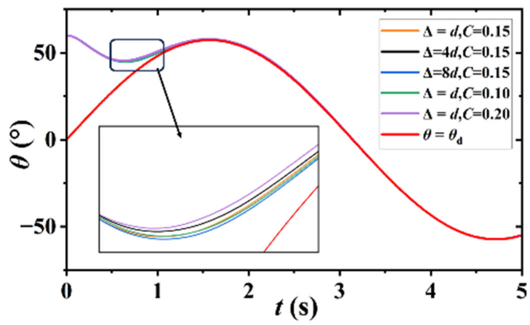

Figure 5. Impact of different Δd and C .

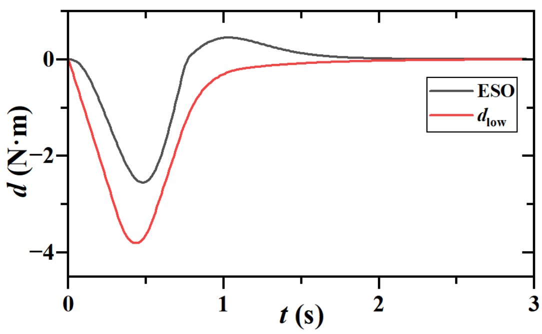

Figure 6. The compensation effect of the ESO.

Analysis of Figure 5 indicates that the introduction of disturbances with varying intensities and the setting of different coupling coefficients exerted a limited influence on the system's tracking accuracy.

Following parameter optimization using PSO, the $H(x^*)$ remained consistently around 0.056. This demonstrates the strong robustness of the proposed system. While disturbances of varying intensity slightly affected the system's convergence time, increasing the coupling coefficient caused a significant variation in the control torque. The system exhibited the longest convergence time among all cases when the coupling coefficient was set to 0.2.

Analysis of Figure 6 reveals that when low-frequency steady-state disturbances were introduced into the system to simulate inaccuracies in estimating the missile's moment of inertia and damping coefficient, the ESO could effectively estimate the magnitude of these disturbances and provide compensation. However, during periods of high disturbance variation rate, the ESO estimation exhibited some deviation. Overall, the performance remained satisfactory. Future work could focus on enhancing the controller by designing a nonlinear or adaptive observer.

The tracking responses of the yaw channel under different initial angles are shown in Figure 7, and the moment variation of the yaw channel is shown in Figure 8.

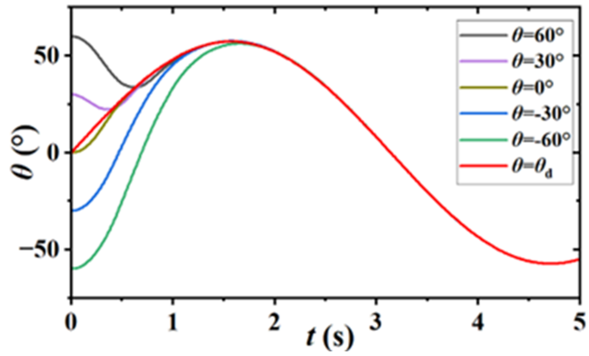

Figure 7. Yaw channel tracking response.

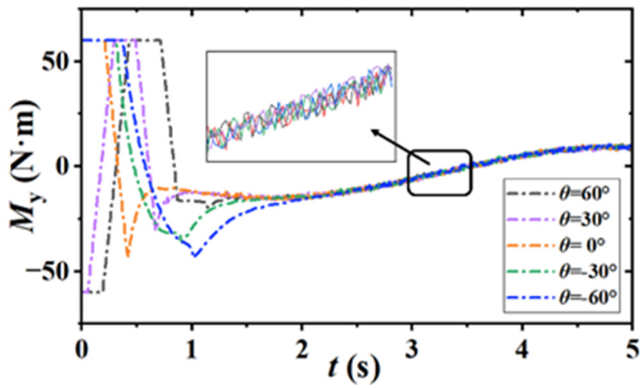

Figure 8. Moment control quantity.

Analysis of Figures 7 and 8 shows that for most initial errors, the system can converge within 1 s, and can follow the target curve well, exhibiting strong disturbance rejection capability. At $\theta = -60°$, the system converges around 1.5 s. Based on the fitness function analysis, under this condition, the large initial error causes the rate of change of the control moment to significantly exceed the prescribed constraint amplitude, resulting in slower convergence. The amplitude of chattering is approximately $1\ \mathrm{N} \cdot m$ and its system chatter is significantly reduced compared to that of traditional sliding mode controllers.

For the yaw channel model of the missile attitude control system, control simulations were conducted by using both a conventional PID controller and a conventional sliding mode controller (SMC). With initial angles set at $-60°$, $-30°$, $30°$, and $60°$, respectively to track the command $\theta_d = \sin t$, the tracking responses are shown in Figures 7 and 8.

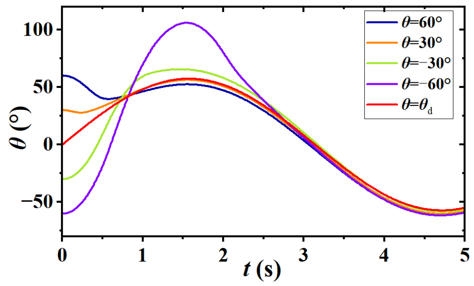

Figure 9. Yaw channel tracking response (PID).

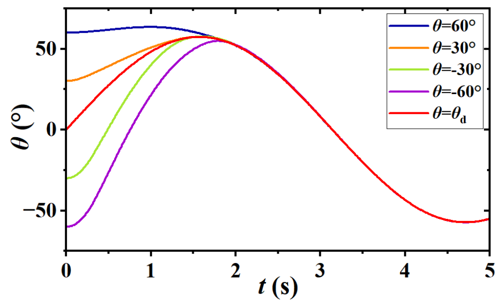

Figure 10. Yaw channel tracking response (SMC).

As shown in Figures 9 and 10: the conventional PID controller exhibits 2-3 s convergence time across different initial angular errors with severe overshoot under large errors and poor tracking performance during rapid command variations; while the traditional SMC achieves ≈2 s convergence, which demonstrates similar initial convergence to NFTSMC (Figure 5) but significantly slower terminal convergence when errors diminish.

6. Conclusion

The PSO-optimized controller demonstrates exceptional comprehensive performance, exhibiting rapid convergence and precise tracking capabilities in simulations. Compared to conventional PID and sliding mode controllers, it achieves an average 35% reduction in convergence time and 40% improvement in control accuracy. The controller design enables enhanced robustness by effectively suppressing aerodynamic disturbances and channel coupling effects. Through global optimization of sliding surface coefficients via PSO, tracking errors are minimized, significantly improving both control precision and system stability.

The ASTA switching control law realizes the dynamic adjustment of gain. This design allows the controller to maintain high gain in areas with large errors to ensure fast response, and reduce gain in areas with small errors to significantly reduce system chatter. Learning mechanism enables the integral term to be adaptive and enhanced according to the persistence of the disturbance, which significantly improves the anti-interference ability of the system.

The integrated control architecture combines NFTSMC, ESO, and PSO, forming a high-efficiency, robust solution framework. Although the proposed method achieves promising results, future work could further enhance performance by optimizing the control moment rate of change while preserving rapid response capabilities and implementing the architecture in high-fidelity 6-DOF models to validate effectiveness in more complex flight scenarios.

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